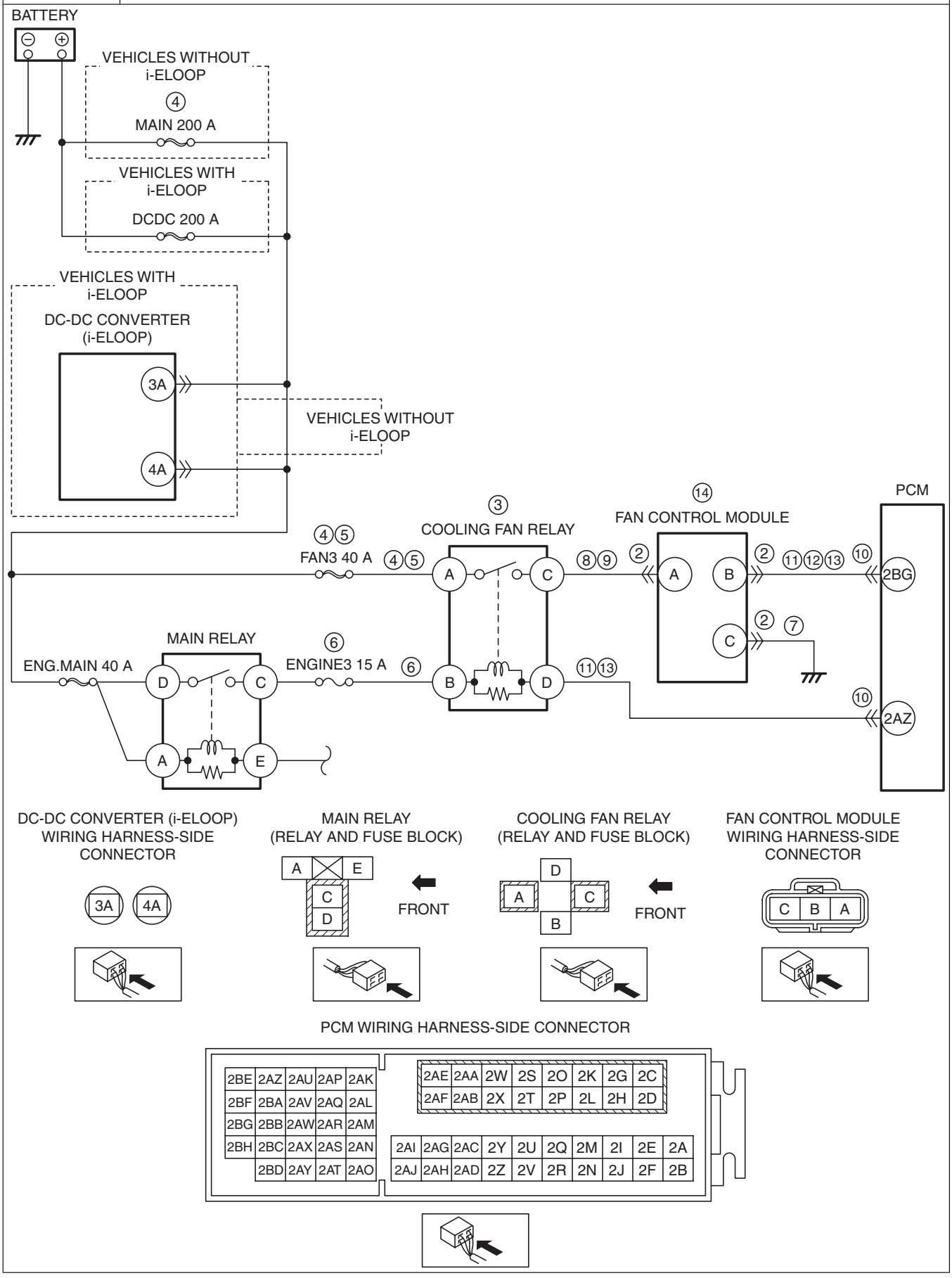
DTC P0480:00 [SKYACTIV-D 1.5]

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DTC P0480:00	Fan control module control circuit problem
DETECTION CONDITION	• The PCM monitors the input voltage from the fan control module. If the voltage at the PCM terminal 2BG remains low or high for **6 s**, the PCM determines that the fan control circuit has a malfunction. **Diagnostic support note** • This is an intermittent monitor (other). • The check engine light does not illuminate. • FREEZE FRAME DATA/Snapshot data is not available. • DTC is not stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Fan control module connector or terminals malfunction • Cooling fan relay malfunction • Short to ground or open circuit in cooling fan relay power supply circuit (without i-ELOOP) — Short to ground in wiring harness between MAIN 200 A fuse and cooling fan relay terminal A — MAIN 200 A fuse malfunction — FAN3 40 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and cooling fan relay terminal A • Short to ground or open circuit in cooling fan relay power supply circuit (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and cooling fan relay terminal A — FAN3 40 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and cooling fan relay terminal A • Short to ground or open circuit in cooling fan relay power supply circuit — Short to ground in wiring harness between ENGINE3 15 A fuse and cooling fan relay terminal B — ENGINE3 15 A fuse malfunction — Open circuit in wiring harness between main relay terminal C and cooling fan relay terminal B • Open circuit in wiring harness between fan control module terminal C and body ground • Short to ground in wiring harness between cooling fan relay terminal C and fan control module terminal A • Open circuit in wiring harness between cooling fan relay terminal C and fan control module terminal A • PCM connector or terminals malfunction • Short to ground in wiring harness between the following terminals: — Fan control module terminal B—PCM terminal 2BG — Cooling fan relay terminal D—PCM terminal 2AZ • Short to power supply in wiring harness between fan control module terminal B and PCM terminal 2BG • Open circuit in wiring harness between the following terminals: — Fan control module terminal B—PCM terminal 2BG — Cooling fan relay terminal D—PCM terminal 2AZ • Fan control module malfunction • PCM malfunction

DTC P0480:00	Fan control module control circuit problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**INSPECT FAN CONTROL MODULE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the fan control module connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 15.
		No	Go to the next step.
3	**INSPECT COOLING FAN RELAY** • Remove the cooling fan relay. (See RELAY LOCATION.) • Inspect the cooling fan relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the cooling fan relay, then go to Step 15.
		No	Vehicles with i-ELOOP: • Go to Step 5. Vehicles without i-ELOOP: • Go to the next step.
4	**INSPECT COOLING FAN RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Cooling fan relay is removed. • Verify that the fan control module connector is disconnected. • Measure the voltage at the cooling fan relay terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to Step 6.
		No	Inspect the MAIN 200 A fuse and FAN3 40 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and cooling fan relay terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and cooling fan relay terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 15.

STEP	INSPECTION		ACTION
5	**INSPECT COOLING FAN RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Cooling fan relay is removed. • Verify that the fan control module connector is disconnected. • Measure the voltage at the cooling fan relay terminal A (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the FAN3 40 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between FAN3 40 A fuse and cooling fan relay terminal A. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which is shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and cooling fan relay terminal A. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Go to Step 15.

STEP	INSPECTION		ACTION
6	**INSPECT COOLING FAN RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Cooling fan relay is removed. • Verify that the fan control module connector is disconnected. • Switch the ignition ON (engine off). • Measure the voltage at the cooling fan relay terminal B (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the ENGINE3 15 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENGINE3 15 A fuse and cooling fan relay terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between main relay terminal C and cooling fan relay terminal B. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 15.
7	**INSPECT FAN CONTROL MODULE GROUND CIRCUIT FOR OPEN CIRCUIT** • Cooling fan relay is removed. • Verify that the fan control module connector is disconnected. • Switch the ignition off. • Inspect for continuity between fan control module terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fan control module terminal C and body ground. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between fan control module and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 15.
8	**INSPECT FAN CONTROL MODULE POWER SUPPLY CIRCUIT FOR SHORT TO GROUND** • Cooling fan relay is removed. • Verify that the fan control module connector is disconnected. • Inspect for continuity between cooling fan relay terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between cooling fan relay terminal C and fan control module terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 15.
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**INSPECT FAN CONTROL MODULE POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Cooling fan relay is removed. • Verify that the fan control module connector is disconnected. • Inspect for continuity between cooling fan relay terminal C (wiring harness-side) and fan control module terminal A (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between cooling fan relay terminal C and fan control module terminal A. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 15.
10	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 15.
		No	Go to the next step.
11	**INSPECT FAN CONTROL MODULE AND COOLING FAN RELAY CIRCUIT FOR SHORT TO GROUND** • Cooling fan relay is removed. • Verify that the fan control module and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Fan control module terminal B — Cooling fan relay terminal D • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Fan control module terminal B—PCM terminal 2BG • Cooling fan relay terminal D—PCM terminal 2AZ **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 15.
		No	Go to the next step.
12	**INSPECT FAN CONTROL MODULE SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Cooling fan relay is removed. • Verify that the fan control module and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the fan control module terminal B (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fan control module terminal B and PCM terminal 2BG. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 15.
13	**INSPECT FAN CONTROL MODULE AND COOLING FAN RELAY CIRCUIT FOR OPEN CIRCUIT** • Cooling fan relay is removed. • Verify that the fan control module and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between the following terminals (wiring harness-side): — Fan control module terminal B—PCM terminal 2BG — Cooling fan relay terminal D—PCM terminal 2AZ • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Fan control module terminal B—PCM terminal 2BG • Cooling fan relay terminal D—PCM terminal 2AZ **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 15.

STEP	INSPECTION		ACTION
14	**INSPECT FAN CONTROL MODULE** • Inspect the fan control module. (See FAN CONTROL MODULE INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the fan control module, then go to the next step. (See COOLING FAN MOTOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.
15	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Perform the KOEO self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
16	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.